

Topic 9: Kinetics I

In order to develop their practical skills, students should be encouraged to carry out a range of practical experiments related to this topic. Possible experiments include investigating a variety of factors that influence the rates of reaction between, for example, marble chips and hydrochloric acid, or sodium thiosulfate and hydrochloric acid, investigating catalysis using hydrogen peroxide.

Mathematical skills that could be developed in this topic include calculating rates from reaction time, plotting graphs and having an appreciation of the nature of the graph for a Maxwell-Boltzmann distribution.

Within this topic, students can consider how the use of models in chemistry is illustrated by the way in which the Maxwell-Boltzmann distribution and collision theory can account for the effects of changing variables on the rate of a chemical reaction.

Students should:
1. understand, in terms of collision theory, the effect of a change in concentration, temperature, pressure and surface area on the rate of a chemical reaction
2. understand that reactions only take place when collisions take place with sufficient energy, known as activation energy
3. be able to calculate the rate of a reaction from: i data showing the time taken for reaction ii the gradient of a suitable graph, by drawing a tangent, either for initial rate, or at a time, t
4. understand qualitatively, in terms of the Maxwell-Boltzmann distribution of molecular energies, how changes in temperature affect the rate of a reaction
5. understand the role of catalysts in providing alternative reaction routes of lower activation energy
6. be able to draw the reaction profiles for uncatalysed and catalysed reactions
7. be able to interpret the action of a catalyst in terms of a qualitative understanding of the Maxwell-Boltzmann distribution of molecular energies
8. understand the use of a solid (heterogeneous) catalyst for industrial reactions, in the gas phase, in terms of providing a surface for the reaction
9. understand the economic benefits of the use of catalysts in industrial reactions